

Week 3, Lecture 2 WOD – Answer Key

A Probability Puzzle

NRES 746 – Instructor Answer Key (not for student distribution)

This WOD is ungraded, so this key isn't a grading rubric – it's a quick reference for circulating during the 5-8 minute window, and a fully worked version of the Part 3 teaser for whenever you reveal it in class.

Part 1: Sevens vs. doubles

1a. No fixed answer – the point is to get a guess on record before counting. Many students guess “sum of 7” is more likely (it's the most commonly known fact about two dice) or “doubles” feels rarer (fewer ways to *name* a double casually). The actual answer (1b-1d) is that they're exactly tied.

1b. $|\Omega| = 36$ (6 outcomes for the red die $\times$ 6 for the blue die, ordered pairs).

1c. Outcomes summing to 7: (1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1) – 6 outcomes. $P(\text{sum} = 7) = 6/36 = 1/6 \approx 0.167$.

1d. Doubles: (1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6) – 6 outcomes. $P(\text{doubles}) = 6/36 = 1/6 \approx 0.167$.

1e. They're **equally likely** – both are exactly 6 of the 36 equally-likely outcomes. Students who guessed “sum of 7” are more likely often reason from partial memory of dice-game lore (7 is famous in craps) without actually enumerating; students who guessed “doubles” often reason that a double feels like a narrower, more special category. Worth pointing out: this only works out to a tie because 7 is the *most-repeated* sum on two dice (six ways to make it, the maximum for any sum) – any other sum (say, a sum of 8, with 5 ways) would actually lose to doubles.

Part 2: How many tickets?

$$\binom{30}{5} = \frac{30!}{5!25!} = \frac{30 \times 29 \times 28 \times 27 \times 26}{5 \times 4 \times 3 \times 2 \times 1}$$

Students only need the expression, not the integer, but for reference:

```
choose(30, 5)
```

```
## [1] 142506
```

= 142,506 tickets. **Common snag:** forgetting to divide by 5! (i.e. answering with the *permutation* count $30 \times 29 \times 28 \times 27 \times 26$ instead of the *combination* count) – that's the “why do you have to divide by something at the end” the hint is pointing at: without dividing, every 5-number ticket gets counted once for each of the $5! = 120$ orders you could have written its numbers down in.

Part 3: The birthday problem

The guess itself has no fixed answer (that's the point of a teaser), but the correct answer, once computed, is **yes** – with 25 people, a shared birthday is *more* likely than not. Most students guess “no,” reasoning (correctly, as far as it goes) that 25 is a small fraction of 365 – the whole reason this is a famous “paradox” is that the intuitive comparison (25 vs. 365) is the wrong comparison to make, as the derivation below shows.

Setting up the calculation

Assume 365 equally likely birthdays (ignore Feb. 29 for simplicity) and that everyone's birthday is independent of everyone else's. Directly computing $P(\text{at least one shared birthday})$ is awkward, because “at least one match” could happen in a huge number of different ways (exactly one pair matches, two separate pairs match, three people share one birthday, etc.). The standard trick: compute the complement instead.

$$P(\text{at least one match}) = 1 - P(\text{no matches at all})$$

“No matches at all” means all 25 birthdays are different – a much easier event to count.

Building “no matches” one person at a time

Line the 25 people up and assign birthdays one at a time, and ask: given everyone assigned so far has a distinct birthday, what's the probability the *next* person also avoids all of them?

- Person 1: no constraint yet – probability of “still all distinct” = $365/365$.
- Person 2: must avoid person 1's one taken birthday – probability = $364/365$.
- Person 3: must avoid the 2 already taken – probability = $363/365$.
- ...
- Person k : must avoid the $(k - 1)$ already-taken birthdays – probability = $(366 - k)/365$.
- ...
- Person 25: must avoid the 24 already taken – probability = $341/365$.

Because each assignment is independent given the ones before it, multiply all 25 probabilities together (this is the same chain-of-conditional-probabilities logic Tuesday's Lab 3 Act 1 used to build up the classifier PPV calculation, and that today's lecture formalizes as Bayes' rule):

$$P(\text{no matches}) = \prod_{k=1}^{25} \frac{366 - k}{365} = \frac{365}{365} \cdot \frac{364}{365} \cdot \frac{363}{365} \cdots \frac{341}{365}$$

Computing it

```
n <- 25
p_no_match <- prod((365 - 0:(n - 1)) / 365)
p_no_match
```

```
## [1] 0.4313003
```

```
p_match <- 1 - p_no_match
p_match
```

```
## [1] 0.5686997
```

$P(\text{no matches}) \approx 0.431$, so $P(\text{at least one match}) \approx 0.569$ – **56.9%**, comfortably past the 50% line.

Why this feels wrong (worth saying out loud in class)

The instinct to compare “25” against “365” is comparing the wrong quantities. The real driver is the number of *pairs* of people, since any pair could be the one that matches:

```
choose(25, 2)
```

```
## [1] 300
```

25 people make $\binom{25}{2} = 300$ distinct pairs. Each individual pair only has a $1/365$ chance of sharing a birthday, but with 300 independent-ish chances to get a match, the odds of *at least one* hit compound quickly – this

is the same “many small chances add up” logic as Act 1’s classifier false-positive rate, just running in the opposite direction (there, many chances to falsely flag; here, many chances to coincidentally match).

A table for context, if useful during the reveal

```
ns <- c(5, 10, 15, 20, 23, 25, 30, 40, 50, 70)
data.frame(
  n = ns,
  P_match = round(1 - sapply(ns, function(n) prod((365 - 0:(n - 1)) / 365)), 3)
)
```

##	n	P_match
## 1	5	0.027
## 2	10	0.117
## 3	15	0.253
## 4	20	0.411
## 5	23	0.507
## 6	25	0.569
## 7	30	0.706
## 8	40	0.891
## 9	50	0.970
## 10	70	0.999

The commonly-cited “tipping point” is $n = 23$ (just past 50%); this WOD deliberately asks about 25 instead so students can’t just recall that number from prior exposure to the classic version of the puzzle.